

Interim Hardware Engineering Report (Weeks 1–5)

1. Executive Summary

This report outlines the key deliverables and engineering milestones achieved during the first five weeks of the hardware engineering internship program. Over this period, the execution of the project has prioritized high-quality work and technical precision over strict timeline adherence. Whether modeling power transient responses, selecting sensor stacks, or executing physical PCB layouts, all outputs have undergone rigorous verification to ensure optimal analog signal integrity, robust power distribution, and overall hardware reliability.

2. Key Deliverables by Week

Week 1: Platform Study & Toolchain Validation

- **Platform Analysis:** Authored a comprehensive comparison matrix for the Fari, Senpai, Sentinel Prime AI, Aido Rover, and Aido Humanoid platforms using public data.
- **Architecture Definition:** Produced a generic subsystem-level block diagram for a mobile service robot, clearly defining compute, sensing, actuation, and power blocks.
- **Environment Setup:** Successfully deployed and verified the complete open-source engineering toolchain, including KiCad 8, LTspice, and PlatformIO.

Week 2: Sensor & Perception Hardware

- **Component Selection:** Developed a sensor selection matrix evaluating over 12 candidate parts tailored for the contrasting use cases of Sentinel Prime AI and Senpai.
- **Schematic Capture & Signal Integrity:** Designed an I2C sensor breakout schematic incorporating IMU, ToF, and I2C buffers. Produced a detailed technical memo analyzing I2C signal integrity, focusing on bus capacitance budgets and pull-up sizing.

Week 3: Power Systems & Battery Management

- **Power Budgets:** Built dual power-budget models for the Aido Rover and Fari platforms, calculating precise active/standby mission times and safe operating margins.
- **BMS & Power Tree Schematic:** Drafted a schematic for a 4S Li-ion BMS protection subsystem alongside a sequenced three-rail power tree (24V → 12V → 5V → 3.3V).
- **Transient Simulation:** Executed an LTspice transient analysis of a 5V buck converter under a 0–2A step load, evaluating overshoot, settling time, and compensation.

Week 4: Embedded Compute & Motor Control

- **MCU Evaluation:** Compiled an MCU comparison matrix assessing candidates for the Aido Rover wheel-motor controller, evaluating critical metrics like ADC speed and deterministic interrupts.
- **Control Implementation:** Wrote a closed-loop velocity controller in PlatformIO for a simulated DC motor plant, complete with a UART telemetry stream and plotted step-response.

Week 5: PCB Physical Design (Mini-Capstone)

- **Fari Sensor Hub Layout:** Leveraged prior schematics to complete a 2-layer, 40 × 40 mm PCB physical design for the Fari sensor hub.

- **Design-for-Manufacture (DFM):** Applied rigorous layout constraints, including dedicated ground pours, precise decoupling capacitor placement, and USB-C ESD protection, resulting in an ERC/DRC-clean board.
- **Manufacturing Outputs:** Generated a complete fabrication bundle (Gerbers, drill files, BOM, CPL) and authored a comprehensive design-decision memo justifying analog routing and component placement choices.

3. Key Learnings & Engineering Growth

The first five weeks have been instrumental in bridging theoretical electrical engineering concepts with practical, system-level robotics hardware design. Key areas of growth include:

- **Translating Schematics to Physical Design:** The Week 5 mini-capstone was a critical exercise in physical layout. Transitioning from ideal schematic representations to a manufacturable 2-layer PCB reinforced the necessity of spatial reasoning, strategic grounding techniques, and strict adherence to DFM rules to ensure a fabrication-ready board.
- **Mixed-Signal Integrity:** Deepened my understanding of practical signal integrity in mixed-signal environments. I learned to mathematically justify I2C pull-up resistor sizing based on total bus capacitance limits rather than relying on default reference values.
- **Power Network Transients:** Gained hands-on experience in modeling power delivery. Simulating buck converters in LTspice highlighted the critical role of proper compensation networks in minimizing voltage overshoot and ensuring stable, clean power rails under transient load steps.
- **Hardware-Software Boundary for Control:** Analyzing MCU architectures specifically for motor control clarified the hardware constraints on firmware. I realized how ADC speed, PWM resolution, and deterministic interrupt latency directly dictate the performance and stability of closed-loop PID controllers.
- **System-Level Trade-offs:** Developed a stronger engineering judgement regarding macro-level product constraints. Anchoring designs to distinct platforms taught me how to balance conflicting requirements, such as engineering a multi-rail power system that meets the high thermal/power envelope of an outdoor Aido Rover while adapting the same principles for the quiet, safety-critical constraints of an indoor Fari unit.

4. Mid-Internship Self-Evaluation

Benchmarked against the Week 9 final evaluation rubric, my current standing is as follows:

- **Technical Depth:** Proficient. Consistently utilized quantitative reasoning to back component selection, particularly during LTspice power simulations and physical PCB design constraints.
- **Tooling Proficiency:** Proficient. Demonstrated fluency across KiCad, LTspice, and embedded compilers, successfully translating theoretical requirements into verified engineering outputs.
- **Design Judgement:** Proficient. Effectively navigated complex trade-offs between contrasting robotic platforms, recognizing the different thermal, acoustic, and computational envelopes of indoor versus outdoor systems.